

Shreya Nair

ML Researcher | AI Interpretability | Causal AI
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Projects

Steering Under Momentum: Affective Switches in Valence-Arousal Space

Completed

Tech Stack: Qwen3-1.7B, PyTorch, HuggingFace, Python, PCA, Ridge Regression | [Paper](#) | [GitHub](#)

- Tested the ability for autoregressive momentum from previous context creation to be able to resist a change in activation steering target in mid-generation within a continuous 2D valence-arousal (VA) subspace by reproducing VA subspace extraction on Qwen3-1.7B with PCA and Ridge Regression on the GoEmotions dataset
- Created a KV-cache branching strategy for generating pairs of continuations on 130 emotion-neutral prompts, 3 switching points, and both valence/arousal dimensions in order to study the causal effects of switching the activation steering target versus natural generation drift
- Created a pull score measure for measuring how well each layer is aligned with the switched activation steering target post-switch, finding that positive valence and low arousal, which RLHF training prefers, are consistently easier to steer into than out of
- Localized the directional and timing biases to the deep third of the model for valence, and found an extra mid-layer separation for arousal, showing that valence and arousal have different representation properties

Explainability using XGBoost for Customer Retention

Completed

Tech Stack: XGBoost, SHAP, AWS SageMaker, Python | [Paper](#) | [GitHub](#)

- Built an explainable AI churn prediction system using Gradient Boosting Methods (XGBoost) and SHAP on a 7,000+ sample dataset, increasing detection of at-risk customers by 50%
- Showed interpretable causal drivers of churn using SHAP feature importance; deployed the model pipeline on AWS SageMaker

Causal Impact Analysis of U.S. Tariff Policy on Indian Exports

Completed

Tech Stack: Causal Inference, Difference-in-Differences, Statsmodels, Linear Regression, Econometrics, Python | [GitHub](#)

- Applied a Difference-in-Differences (DiD) causal inference framework to isolate the treatment effect of U.S. tariff policy on Indian export sectors, comparing affected industries against a matched control group
- Built an econometric pipeline using OLS regression and Statsmodels to estimate treatment effects and validate the parallel-trends assumption across pre- and post-tariff periods
- Found a statistically insignificant treatment effect ($p = 0.697$), showing the estimated \$23.23M export decline was indistinguishable from random economic noise rather than a genuine tariff impact

Work Experience

VIT Vellore

Undergraduate Researcher

Vellore, IN

Dec 2025 – May 2026

Under the guidance of Dr. Monica Subhashini

- Applied the GFCI algorithm for Causal Discovery Analysis to identify causal mechanisms in Autism diagnosis by integrating fMRI, sMRI, and phenotypic data
- Built a multimodal deep learning pipeline inspired by GFCI, a hybrid of GES (Greedy Equivalence Search) and FCI (Fast Causal Inference), to fuse neuroimaging and behavioral features for robust causal discovery beyond purely statistical approaches

Grant Thornton Bharat

Data Analyst Intern

Bengaluru, IN

Jun 2026 – Present

- Conducting data analysis using SAP Analytics Cloud (SAC) and SQL to surface actionable business insights across client engagements

WorldQuant

Research Consultant

Remote

Aug 2025 – Present

- Developed mathematical models for quantitative investment strategies; optimized trading alphas achieving Sharpe (~2.5) and fitness (~1.5) through rigorous backtesting

Education

Vellore Institute of Technology

B.Tech in Electronics and Instrumentation

Vellore, TN

Expected April 2027

CGPA: 8.94/10 | Branch Topper 2024

- Relevant Coursework and Certifications: Machine Learning, Artificial Intelligence, Statistics.

Technical Skills

Languages: Python, MATLAB, SQL, Java, Git

ML / DL Tools: PyTorch, TensorFlow, TransformerLens, HuggingFace, Scikit-Learn, SHAP, Statsmodels

Data & Cloud: Pandas, NumPy, Matplotlib, Seaborn, SAP Analytics Cloud (SAC), AWS (SageMaker, Lambda, SNS)

Concepts: AI Interpretability, Causal Inference, Explainable AI, Statistical Analysis, NLP, Deep Learning

Achievements

- Women Achiever of the Year (2026), Recognised by VIT for outstanding performance and leadership across MUN collegiate circuits from 2024 to 2025
- Branch Topper (2024) and Dean's Merit List (2024 & 2025), Recognised for academic excellence in Electronics and Instrumentation Engineering